Document 1: Theoretical Setup for Bartlett-Based Correction Methods in Structural Equation Modeling

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A comprehensive treatment of the theoretical foundations underlying Bartlett-type corrections for the likelihood ratio test in covariance structure analysis

Contents

1 Introduction and Scope

The likelihood ratio test (LRT) is one of the most widely used methods for testing hypotheses about covariance structures in multivariate statistics and structural equation modeling (SEM). Under standard regularity conditions, the LRT statistic converges in distribution to a chi-square random variable as the sample size grows. However, this asymptotic approximation can be markedly inaccurate in finite samples, especially when the number of observed variables is large relative to the sample size, or when the number of free parameters approaches the total number of available statistics. This discrepancy between the nominal and actual Type I error rates has motivated the development of correction methods that adjust the test statistic to better approximate the limiting chi-square distribution at finite sample sizes.

The foundational work on this problem was published by Bartlett (1937), who demonstrated that the expected value of the likelihood ratio statistic under the null hypothesis differs from the nominal degrees of freedom by a term of order $O(n^{-1})$, where n is the sample size. Bartlett proposed dividing the test statistic by a correction factor involving the sample size and the number of parameters, thereby improving the chi-square approximation. This simple yet powerful idea has spawned a rich literature spanning nearly nine decades, with adaptations and extensions tailored to specific model classes and data conditions.

The historical hierarchy of Bartlett-type corrections relevant to SEM can be summarized as follows. Bartlett's original 1937 paper established the general framework for correcting the LRT. Bartlett (1950) applied the correction specifically to factor analysis models, deriving an explicit correction factor that depends on the number of observed variables and the number of latent factors. Building upon this, Swain (1975) derived a general correction for covariance structure models estimated by maximum likelihood, providing a formula that accounts for both the number of observed variables and the number of free parameters. Most recently, Yuan (2015) proposed an empirical correction that adjusts the degrees of freedom rather than the test statistic itself, demonstrating improved performance under conditions of non-normality. Throughout this hierarchy, maximum likelihood (ML) estimation serves as the baseline methodology against which all corrections are evaluated.

This document provides a self-contained and comprehensive treatment of the theoretical foundations underlying all Bartlett-type correction methods for the LRT in SEM. We begin with the general hypothesis testing framework and the standard LRT statistic, proceed through the maximum likelihood fitting function and its properties, and then develop the Bartlett correction concept in full generality. We present detailed derivations of each correction method, including the uncorrected ML test (as a baseline), Swain's correction, Bartlett's EFA correction, and Yuan's empirical correction. We also discuss the complexity matrix that motivates our simulation design, Bradley's robustness criterion for evaluating Type I error rates, and the impact of non-normality on the validity of Bartlett corrections.

2 The Likelihood Ratio Test in SEM

2.1 Hypothesis Testing Framework

In structural equation modeling, the central inferential question concerns whether a hypothesized covariance structure adequately reproduces the population covariance matrix. Let $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n$ denote a random sample of size n from a p-variate distribution with mean vector $\boldsymbol{\mu}$ and positive definite covariance matrix $\boldsymbol{\Sigma}$. The model specifies that $\boldsymbol{\Sigma}$ has a particular parametric form $\boldsymbol{\Sigma}(\boldsymbol{\theta})$, where $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^t$ is a t-dimensional vector of free parameters. The null hypothesis of interest is:

$$H_0\colon \boldsymbol{\Sigma} = \boldsymbol{\Sigma}(\boldsymbol{\theta}_0), \quad \text{for some } \boldsymbol{\theta}_0 \in \Theta, \tag{1}$$

which asserts that the population covariance matrix belongs to the family of matrices generated by the model. The alternative hypothesis is that $\boldsymbol{\Sigma}$ is an arbitrary positive definite matrix, unrestricted by the model. Under the alternative, the number of free parameters is $p(p+1)/2$, corresponding to the number of unique elements of $\boldsymbol{\Sigma}$ (since $\boldsymbol{\Sigma}$ is symmetric). Under the null, the number of free parameters is t, the dimensionality of $\boldsymbol{\theta}$. The degrees of freedom for the test are therefore:

$$\mathrm{df} = \frac{p(p+1)}{2} - t. \tag{2}$$

A necessary condition for the model to be identified and testable is that $\mathrm{df} > 0$, meaning that the number of available statistics $p(p+1)/2$ must strictly exceed the number of free parameters t. When $\mathrm{df} = 0$, the model is said to be *just-identified* or *saturated*, and the test has no power. When $\mathrm{df} < 0$, the model is *under-identified* and cannot be estimated.

2.2 The LRT Statistic

The likelihood ratio test statistic is constructed from the ratio of the maximized likelihood under the null hypothesis to the maximized likelihood under the alternative. Under multivariate normality, the log-likelihood function (up to an additive constant not depending on the parameters) is proportional to:

$$\ell(\boldsymbol{\theta}) = -\frac{n}{2}\big[\ln|\boldsymbol{\Sigma}(\boldsymbol{\theta})| + \mathrm{tr}\big(\mathbf{S}\,\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\big)\big], \tag{3}$$

where $\mathbf{S} = n^{-1}\sum_{i=1}^{n}(\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top$ is the sample covariance matrix. Under the alternative hypothesis, $\boldsymbol{\Sigma}$ is unrestricted, and the maximum likelihood estimator is $\hat{\boldsymbol{\Sigma}}_{\text{alt}} = \mathbf{S}$ (up to the factor of $(n-1)/n$, which is negligible for large n). Under the null hypothesis, we maximize $\ell(\boldsymbol{\theta})$ over $\boldsymbol{\theta} \in \Theta$ to obtain $\hat{\boldsymbol{\theta}}$.

The likelihood ratio statistic is defined as:

$$w = -2\ln\Lambda = -2\big[\ell(\hat{\boldsymbol{\theta}}) - \ell(\hat{\boldsymbol{\Sigma}}_{\text{alt}})\big]. \tag{4}$$

After simplification, this reduces to:

$$w = n\big[\ln|\mathbf{\Sigma}(\hat{\boldsymbol{\theta}})| + \mathrm{tr}\big(\mathbf{S}\,\mathbf{\Sigma}(\hat{\boldsymbol{\theta}})^{-1}\big) - \ln|\mathbf{S}| - p\big]. \tag{5}$$

Note that $w \geq 0$, with equality if and only if $\mathbf{S} = \mathbf{\Sigma}(\hat{\boldsymbol{\theta}})$ (i.e., the model perfectly reproduces the sample covariance matrix).

2.3 Wilks' Theorem

The fundamental asymptotic result underpinning the use of the LRT is *Wilks' theorem* (Wilks, 1938), which states that under regularity conditions—including that the null hypothesis lies in the interior of the parameter space, that the Fisher information matrix is positive definite at $\boldsymbol{\theta}_0$, and that the observations are independently and identically distributed—the LRT statistic converges in distribution to a chi-square random variable:

$$w \xrightarrow{d} \chi^2_{\mathrm{df}} \quad \text{as } n \to \infty, \tag{6}$$

where $\mathrm{df} = p(p+1)/2 - t$ is given by Equation (2). The convergence rate is typically $O(n^{-1/2})$, meaning that the quality of the chi-square approximation improves as the sample size increases. However, in practice, the quality of the approximation also depends on the ratio of p to n and the complexity of the model, as we discuss in subsequent sections.

It is important to emphasize that Wilks' theorem provides only a *first-order* asymptotic result. The cumulative distribution function of w is approximated by the chi-square distribution to order $O(n^{-1})$, meaning that the difference between the true distribution of w and the reference χ^2_{df} distribution decays at the rate $n^{-1/2}$ for the distribution function, but the error in the tail probabilities (which are of primary interest in hypothesis testing) can be substantially larger. This observation is precisely what motivates the Bartlett correction.

3 The Maximum Likelihood Fitting Function

3.1 The Wishart Distribution

Under multivariate normality, the sufficient statistic for $\mathbf{\Sigma}$ is the sample covariance matrix $\mathbf{S}$. The sampling distribution of $\mathbf{S}$ is given by the Wishart distribution. Specifically, let $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n)$ be an $n \times p$ data matrix, and define the (bias-corrected) sample covariance matrix as:

$$\mathbf{S}^* = \frac{1}{n-1}\sum_{i=1}^{n}(\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top. \tag{7}$$

Then, when the population distribution is $N_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, the matrix $(n-1)\mathbf{S}^*$ follows the Wishart distribution with $n-1$ degrees of freedom and scale matrix $\mathbf{\Sigma}$:

$$(n-1)\mathbf{S}^* \sim W_p(n-1,\, \mathbf{\Sigma}). \tag{8}$$

The Wishart distribution is the multivariate generalization of the gamma distribution and plays a central role in multivariate analysis. The expected value and variance of the elements of $\mathbf{S}^*$

under the Wishart distribution are:

$$\mathbb{E}[\mathbf{S}^*] = \boldsymbol{\Sigma}, \tag{9}$$

$$\mathrm{Var}[S^*_{ij}] = \frac{\sigma_{ii}\sigma_{jj} + \sigma_{ij}^2}{n-1}, \tag{10}$$

where S^*_{ij} and σ_{ij} denote the (i, j)-th elements of $\mathbf{S}^*$ and $\boldsymbol{\Sigma}$, respectively. These expressions reveal that the variability of the sample covariance matrix decreases as the sample size increases, with the rate of convergence being $O(n^{-1/2})$ for each element.

3.2 The ML Fitting Function

The maximum likelihood fitting function for covariance structure models is derived by minimizing the Kullback–Leibler divergence between the sample covariance matrix $\mathbf{S}$ and the model-implied covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\theta})$. Under multivariate normality, this minimization is equivalent to maximizing the log-likelihood function. The ML fitting function, often denoted F_{ML}, is defined as:

$$F_{\mathrm{ML}}(\boldsymbol{\theta}) = \ln|\boldsymbol{\Sigma}(\boldsymbol{\theta})| + \mathrm{tr}\big(\mathbf{S}\,\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}\big) - \ln|\mathbf{S}| - p, \tag{11}$$

where $\mathbf{S} = n^{-1}\sum_{i=1}^{n}(\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top$ is the MLE of $\boldsymbol{\Sigma}$ under the unrestricted model. The function F_{ML} has several important properties. First, it is non-negative for all $\boldsymbol{\theta} \in \Theta$, with $F_{\mathrm{ML}}(\boldsymbol{\theta}) = 0$ if and only if $\mathbf{S} = \boldsymbol{\Sigma}(\boldsymbol{\theta})$. Second, F_{ML} is a discrepancy function in the sense of Browne (1974), meaning it satisfies the conditions of a proper statistical distance measure between $\mathbf{S}$ and $\boldsymbol{\Sigma}(\boldsymbol{\theta})$. Third, F_{ML} is invariant under simultaneous rescaling and rotation of the observed variables.

3.3 The ML Test Statistic

The standard ML test statistic is obtained by multiplying the minimized value of the fitting function by the sample size:

$$T_{\mathrm{ML}} = n \cdot F_{\mathrm{ML}}(\hat{\boldsymbol{\theta}}), \tag{12}$$

where $\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} F_{\mathrm{ML}}(\boldsymbol{\theta})$ is the ML estimator of $\boldsymbol{\theta}$. Under the null hypothesis and multivariate normality, T_{ML} is asymptotically distributed as χ^2_{df}, where df is given by Equation (2). In practice, a commonly used variant replaces n with $n-1$:

$$T_{\mathrm{ML}} = (n-1) \cdot F_{\mathrm{ML}}(\hat{\boldsymbol{\theta}}), \tag{13}$$

which provides a small-sample correction that partially accounts for the loss of one degree of freedom due to estimation of the mean vector. The difference between using n and $n-1$ is negligible for large samples but can be noticeable when n is small relative to p.

4 The Bartlett Correction Concept

4.1 Bartlett's Original Idea (1937)

The key insight of Bartlett (1937) was that the chi-square approximation to the distribution of the LRT statistic can be systematically improved by adjusting the statistic to account for the $O(n^{-1})$ bias in its expected value. Specifically, Bartlett showed that under the null hypothesis, the expected value of w admits an asymptotic expansion of the form:

$$\mathbb{E}[w] = \mathrm{df} + \frac{\varepsilon}{n} + O(n^{-2}), \tag{14}$$

where ε is a constant that depends on the model and the dimensionality of the data. Since the expected value of a χ^2_{df} random variable is exactly df, the discrepancy ε/n represents the leading-order bias in the LRT statistic. When $\varepsilon > 0$, the uncorrected LRT statistic tends to be too large on average, leading to inflated Type I error rates (the test rejects the null hypothesis too often). When $\varepsilon < 0$, the opposite occurs.

Bartlett proposed correcting the test statistic by dividing it by a factor that removes the leading-order bias:

$$w^* = \frac{w}{1 + b/n}, \tag{15}$$

where $b = \varepsilon/\mathrm{df}$ is the Bartlett correction factor. Equivalently, one may write:

$$w^* = \frac{w}{c}, \qquad c = 1 + \frac{b}{n}, \tag{16}$$

where c is the Bartlett correction factor. The corrected statistic w^* satisfies:

$$\mathbb{E}[w^*] = \mathrm{df} + O(n^{-2}), \tag{17}$$

so that the bias has been reduced from $O(n^{-1})$ to $O(n^{-2})$. This improvement translates to better agreement between the empirical distribution of the test statistic and the reference χ^2_{df} distribution, as demonstrated by the following theorem of Bartlett.

Bartlett (1937) — Improved Chi-Square Approximation

Let $w^* = w/(1 + b/n)$, where $b = \varepsilon/\mathrm{df}$ and ε is defined in Equation (14). Then:

1. The distribution of w^* agrees with that of χ^2_{df} to order $O(n^{-2})$ in the sense of Edgeworth expansion matching.
2. Specifically, the cumulative distribution functions satisfy:

$$\Pr(w^* \leq x) = \Pr(\chi^2_{\mathrm{df}} \leq x) + O(n^{-2}). \tag{18}$$

This result is remarkable because it shows that a simple multiplicative correction, requiring only the computation of a single constant b, can improve the accuracy of the chi-square approximation by a full order of magnitude. The practical implication is that for moderate sample sizes, the corrected statistic w^* provides substantially more accurate p-values than the uncorrected statistic w.

4.2 Lawley's General Formula (1956)

The general formula for the bias term ε was derived by Lawley (1956) in the context of testing hypotheses about covariance matrices. Lawley showed that ε can be expressed in terms of the fourth-order cumulants of the sufficient statistics under the null hypothesis. For the case of covariance structure testing, the relevant cumulants are those of the sample covariance matrix under the Wishart distribution.

Let Λ_{rstu} denote the fourth-order cumulant of the sample covariance matrix elements, defined as:

$$\Lambda_{rstu} = \kappa\big(S_{rs},\, S_{tu}\big) = \mathbb{E}[S_{rs}\, S_{tu}] - \mathbb{E}[S_{rs}]\, \mathbb{E}[S_{tu}], \tag{19}$$

where S_{rs} denotes the (r, s)-th element of the sample covariance matrix $\mathbf{S}$. Under the Wishart distribution $W_p(n, \mathbf{\Sigma})$, these cumulants have a well-known closed-form expression:

$$\Lambda_{rstu} = \frac{1}{n}\big(\sigma_{rs}\,\sigma_{tu} + \sigma_{rt}\,\sigma_{su} + \sigma_{ru}\,\sigma_{st}\big), \tag{20}$$

where σ_{rs} denotes the (r, s)-th element of $\mathbf{\Sigma}$. Lawley's formula expresses the bias term as a sum involving products of these cumulants and the second derivatives of the fitting function with respect to the model parameters.

4.3 Lawley's Theorem

A deep result in the theory of Bartlett corrections, due to Lawley (1956), states that when the Bartlett correction is applied, not only does the expected value of the corrected statistic agree with df to a higher order, but the entire distribution—including all cumulants—agrees with the chi-square distribution to order $O(n^{-3/2})$.

Lawley (1956) — Cumulant Matching

Let $w^* = w/(1 + b/n)$ be the Bartlett-corrected LRT statistic with the optimal correction factor b. Then, for $r = 1, 2, \ldots$:

1. The r-th cumulant of w^* satisfies:

$$\kappa_r(w^*) = \kappa_r(\chi^2_{\mathrm{df}}) + O(n^{-3/2}), \tag{21}$$

2. where $\kappa_r(\chi^2_{\mathrm{df}}) = 2^{r-1}(r-1)!\,\mathrm{df}$ is the r-th cumulant of the chi-square distribution with df degrees of freedom.

This theorem provides a powerful theoretical justification for the Bartlett correction: by matching the first cumulant (the mean) to a higher order, one automatically achieves improved agreement for all higher-order cumulants. The proof relies on the structure of the Edgeworth expansion and the fact that the Bartlett correction removes the leading-order term in the expansion that distinguishes the distribution of w from the chi-square distribution.

5 Complexity Matrix: Model Design Based on Comparative Weight of p and t

5.1 The Identification Constraint

A fundamental constraint in structural equation modeling is that the model must be identified, which requires that the number of unique elements in the covariance matrix, $p(p+1)/2$, exceeds the number of free parameters t. The degrees of freedom for the chi-square test are:

$$\mathrm{df} = \frac{p(p+1)}{2} - t > 0. \tag{22}$$

The ratio of free parameters to available statistics, which we term the *complexity ratio*, is defined as:

$$\rho = \frac{t}{p(p+1)/2}. \tag{23}$$

This ratio determines how much of the available information (i.e., the $p(p+1)/2$ unique variance and covariance elements) is consumed by the model parameters. When ρ is small, the model is parsimonious and the test has many degrees of freedom, which generally leads to a well-behaved chi-square approximation. When ρ approaches 1, the model is nearly saturated, the degrees of freedom are few, and the chi-square approximation tends to be poor. The Bartlett correction is most needed—and most effective—in intermediate regimes where the uncorrected approximation is inadequate but the model is not so saturated as to render any finite-sample correction unreliable.

5.2 Three-Model Design

To systematically investigate the performance of Bartlett corrections across different levels of model complexity, we design three models that span a range of relationships between p and t. The design criteria are as follows: each model should represent a realistic SEM application, the degrees of freedom should be large enough to permit meaningful hypothesis testing, and the three models should differ substantially in their complexity ratios to reveal how the Bartlett correction interacts with model parsimony.

5.3 Model 1: Simple CFA ($p = 9$, $t = 21$, df$= 24$)

Model 1 is a confirmatory factor analysis model with $p = 9$ observed variables organized into $q = 3$ latent factors, each measured by 3 observed indicators. The free parameters include 9 factor loadings, 9 error variances, and 3 factor covariances, yielding $t = 21$ free parameters. The number of unique covariance elements is $p(p + 1)/2 = 45$, so the degrees of freedom are $\mathrm{df} = 45 - 21 = 24$.

Table 1: Summary of the three simulation models and their characteristics.

Characteristic	**Model 1**	**Model 2**	**Model 3**
Model Type	Simple CFA	Medium CFA	Full SEM
Observed vars (p)	9	12	9
Free parameters (t)	21	27	22
Unique elements ($p(p{+}1)/2$)	45	78	45
Degrees of freedom	24	51	23
Complexity ratio (ρ)	0.467	0.346	0.489

In this model, the complexity ratio is $\rho = 21/45 \approx 0.467$, meaning that approximately 46.7% of the available information is consumed by the model parameters. The remaining 53.3% of the information budget provides the basis for the chi-square test. This model represents a moderately parsimonious specification in which the chi-square approximation is expected to be reasonably good for large n but may exhibit noticeable bias for moderate sample sizes (e.g., $n < 200$). The relatively large degrees of freedom (df = 24) ensure that the chi-square reference distribution is well-approximated by a normal distribution, which aids the convergence of the Bartlett correction.

5.4 Model 2: Medium CFA ($p = 12$, $t = 27$, df$= 51$)

Model 2 is a larger confirmatory factor analysis model with $p = 12$ observed variables and $q = 3$ latent factors, each measured by 4 observed indicators. The free parameters include 12 factor loadings, 12 error variances, and 3 factor covariances, yielding $t = 27$ free parameters. The number of unique covariance elements is $p(p + 1)/2 = 78$, so the degrees of freedom are $\mathrm{df} = 78 - 27 = 51$.

This model has a substantially lower complexity ratio of $\rho = 27/78 \approx 0.346$, reflecting a more information-rich specification in which the model is relatively parsimonious compared to the number of available statistics. The large degrees of freedom (df = 51) mean that the chi-square reference distribution is very well approximated by a normal distribution, and the Bartlett correction is expected to have a modest effect for large n. However, for smaller sample sizes and non-normal data, the correction may still provide meaningful improvements in Type I error control. This model serves as a benchmark for conditions under which the uncorrected LRT is already reasonably accurate, allowing us to assess whether the Bartlett correction maintains adequate performance (i.e., does not over-correct) in favorable conditions.

5.5 Model 3: Full SEM ($p = 9$, $t = 22$, df$= 23$)

Model 3 is a full structural equation model with $p = 9$ observed variables. Unlike Models 1 and 2, this model includes both a measurement component (confirmatory factor analysis) and a structural component (regression paths among latent variables). The total number of free parameters is $t = 22$, resulting in degrees of freedom $\mathrm{df} = 45 - 22 = 23$.

The complexity ratio is $\rho = 22/45 \approx 0.489$, which is the highest among the three models. This near-saturated specification means that the model consumes nearly half of the available information, leaving relatively few degrees of freedom for hypothesis testing. In such a configuration, the chi-square approximation to the distribution of the LRT statistic is known to be particularly poor, especially for small sample sizes. The Bartlett correction is expected to have its largest effect in this model, as the $O(n^{-1})$ bias term is amplified when the degrees of freedom are few relative to the number of parameters. Additionally, the inclusion of structural paths introduces additional complications for the Bartlett correction, as the formula must account for the dependencies between the measurement and structural components of the model.

5.6 Impact of the Complexity Ratio on the Bartlett Correction

The complexity ratio $\rho = t/[p(p+1)/2]$ directly influences the magnitude of the Bartlett correction factor. Intuitively, models with higher complexity ratios have less information available for testing, and the finite-sample bias of the LRT statistic tends to be larger. In the Swain (1975) correction, the bias term is proportional to $t + p$, which grows with both the number of parameters and the number of variables. In the Bartlett (1950) EFA correction, the bias term depends on $p + 2q$, reflecting the role of both the dimensionality and the number of factors. The Yuan (2015) empirical correction adjusts the degrees of freedom based on the estimated bias, which is also influenced by the complexity ratio. By spanning a range of complexity ratios from 0.346 to 0.489, our three-model design allows us to investigate these relationships empirically.

6 Correction Methods: Detailed Derivations

This section presents the detailed derivations of all four correction methods considered in this study. Each method is presented with its mathematical foundations, practical formula, and key properties.

6.1 Uncorrected ML (Baseline)

The uncorrected ML test statistic serves as the baseline against which all correction methods are evaluated. It is defined as:

$$T_{\mathrm{ML}} = (n-1) \cdot F_{\mathrm{ML}}(\hat{\boldsymbol{\theta}}), \tag{24}$$

where $\hat{\boldsymbol{\theta}}$ is the ML estimator and F_{ML} is the ML fitting function given in Equation (11). Under the null hypothesis and multivariate normality, T_{ML} converges in distribution to χ^2_{df} as $n \to \infty$.

The uncorrected statistic is included as a reference for several important reasons. First, it provides the simplest and most widely used test, and its performance characteristics are well understood. Second, the magnitude of the Bartlett correction can be assessed by comparing the corrected statistics to T_{ML}: a large correction indicates that the uncorrected statistic is substantially biased. Third, the simulation results for T_{ML} establish a benchmark for Type I error rates; any correction method should maintain or improve upon these rates. Fourth, under non-normality, T_{ML} is known to over-reject (i.e., the empirical Type I error rate exceeds the nominal level), and the degree of over-rejection provides a measure of the sensitivity of the test to distributional assumptions.

A known property of T_{ML} is that its expected value under the null differs from df by a term of order $O(n^{-1})$. Specifically, for covariance structure models estimated by ML:

$$\mathbb{E}[T_{\mathrm{ML}}] = \mathrm{df} + \frac{2t + 2p + 3}{2n} + O(n^{-2}), \tag{25}$$

where t is the number of free parameters and p is the number of observed variables. This expression reveals that the bias is positive and proportional to $(t + p + 1.5)/n$, confirming that the uncorrected statistic tends to be too large on average. The bias increases with both the number of parameters and the number of variables, and decreases with the sample size. This expression also provides the theoretical basis for the Swain correction and the Yuan correction, as discussed below.

6.2 Swain's Correction (1975)

Swain's Correction Factor

The Swain (1975) correction factor is defined as:

$$c_{\mathrm{Swain}} = \frac{2n - 2t - 2p - 1}{2t + 2p + 3}, \tag{26}$$

and the corrected test statistic is:

$$T_{\mathrm{Swain}} = T_{\mathrm{ML}} \cdot \frac{2t + 2p + 3}{2n - 2t - 2p - 1}. \tag{27}$$

Swain (1975) derived this correction by applying Lawley's (1956) general formula for the Bartlett correction to the specific case of covariance structure models estimated by maximum likelihood. The key steps in the derivation are as follows.

Step 1: Express the bias in terms of model characteristics. From Lawley's general result, the bias in the LRT statistic under the null hypothesis for covariance structure models is:

$$\varepsilon_{\mathrm{Swain}} = \frac{2t + 2p + 3}{2}, \tag{28}$$

where t is the number of free parameters and p is the number of observed variables. This expression arises from evaluating the sum of products of second derivatives of the fitting function and fourth-order cumulants of the sample covariance matrix under the Wishart distribution. The detailed derivation involves summing over all parameter indices and all pairs of unique covariance elements, which yields the compact expression above.

Step 2: Construct the Bartlett correction factor. The Bartlett correction factor is:

$$1 + \frac{b}{n} = 1 + \frac{\varepsilon_{\mathrm{Swain}}}{n \cdot \mathrm{df}} = \frac{n + \varepsilon_{\mathrm{Swain}}/\mathrm{df}}{n}. \tag{29}$$

Swain's correction uses a slightly different parameterization. Instead of directly dividing by

$1 + b/n$, Swain observes that:

$$\frac{T_{\mathrm{ML}}}{1 + (2t + 2p + 3)/(2n)} \approx T_{\mathrm{ML}} \cdot \frac{2n}{2n + 2t + 2p + 3}. \tag{30}$$

After accounting for the factor $(n-1)$ used in the definition of T_{ML} (rather than n), the corrected statistic becomes:

$$T_{\mathrm{Swain}} = T_{\mathrm{ML}} \cdot \frac{2t + 2p + 3}{2n - 2t - 2p - 1}, \tag{31}$$

where we have used the identity $2(n - 1) = 2n - 2$ and simplified.

Step 3: Properties and limitations. Swain's correction has several important properties. First, the correction factor is always positive when $2n - 2t - 2p - 1 > 0$, which requires:

$$n > t + p + 0.5. \tag{32}$$

This condition is typically satisfied in practice, since identification requires $p(p + 1)/2 > t$ and adequate sample sizes require $n \gg p$. However, when the sample size is small relative to $t+p$, the correction factor can become very large, potentially making the corrected statistic unrealistically small. Second, the correction always reduces the value of the test statistic (since the multiplier $(2t+2p+3)/(2n-2t-2p-1)$ is typically less than 1 for n moderately large), which is consistent with the positive bias of the uncorrected statistic. Third, the correction depends on both p and t, reflecting the fact that the bias increases with the dimensionality of the model.

Negative Correction Factor Warning

When $2n - 2t - 2p - 1 \leq 0$, the Swain correction factor becomes non-positive, and the corrected statistic is undefined or takes a nonsensical value. This occurs when $n \leq t + p + 0.5$. In practice, this means that Swain's correction should not be applied to very small samples or very complex models. For the three models in our simulation design, the minimum sample sizes required for a positive correction factor are: Model 1 ($n > 30.5$), Model 2 ($n > 39.5$), Model 3 ($n > 31.5$).

6.3 Bartlett EFA Correction (1950)

Bartlett's EFA Correction Factor

For an exploratory factor analysis model with p observed variables and q factors, Bartlett (1950) proposed the correction factor:

$$c_{\mathrm{Bart}} = \frac{n - 1 - (2p + 4q + 1)/2}{n - 1} = \frac{2n - 2p - 4q - 3}{2n - 2}, \tag{33}$$

and the corrected test statistic is:

$$T_{\mathrm{Bart}} = \frac{T_{\mathrm{ML}}}{c_{\mathrm{Bart}}} = T_{\mathrm{ML}} \cdot \frac{2n - 2}{2n - 2p - 4q - 3}. \tag{34}$$

Bartlett's (1950) correction was derived specifically for exploratory factor analysis models, in which the model is:

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\,\boldsymbol{\Phi}\,\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \tag{35}$$

where $\boldsymbol{\Lambda}$ is the $p \times q$ matrix of factor loadings, $\boldsymbol{\Phi}$ is the $q \times q$ factor covariance matrix (typically constrained to be the identity matrix in EFA for rotation identification), and $\boldsymbol{\Psi}$ is the $p \times p$ diagonal matrix of unique variances.

Derivation. Bartlett's derivation proceeds by counting the effective number of parameters in the EFA model and then applying the general Bartlett correction framework. In EFA with q factors, the number of free parameters is:

$$t_{\text{EFA}} = p(q+1) - \frac{q(q-1)}{2}, \tag{36}$$

where pq parameters are the factor loadings, p parameters are the unique variances, and the term $q(q-1)/2$ accounts for the rotational indeterminacy of the factor loading matrix (which reduces the effective number of free parameters). However, Bartlett's correction does not use this exact parameter count. Instead, Bartlett (1950) showed that the bias term for the EFA model is:

$$\varepsilon_{\text{Bart}} = \frac{2p + 4q + 1}{2}, \tag{37}$$

This expression differs from Swain's bias term $\varepsilon_{\text{Swain}} = (2t + 2p + 3)/2$ in a crucial way: the Bartlett EFA bias depends on p and q (the number of factors) rather than on the total number of free parameters t. This is because the rotational indeterminacy of the EFA model means that not all t parameters contribute equally to the bias; some parameters are redundant due to the factor rotation invariance, and their contribution to the bias cancels out in the derivation.

Original form. Bartlett (1950) originally presented the correction in the form:

$$T_{\text{Bart}} = \frac{T_{\text{ML}}}{1 - (2p + 4q + 1)/(2n - 2)}, \tag{38}$$

which is algebraically equivalent to Equation (34). This form makes it clear that the correction factor $c_{\text{Bart}} < 1$ when $(2p + 4q + 1) > 0$, which is always the case in practice. The denominator $1 - (2p + 4q + 1)/(2n - 2)$ decreases as p or q increases, and increases as n increases, consistent with the intuition that the correction is most needed for complex models with small sample sizes.

Comparison with Swain's correction. The key differences between Bartlett's EFA correction and Swain's correction are:

1. **Parameter dependence:** Bartlett's correction depends on p and q (number of factors), while Swain's correction depends on p and t (total number of free parameters). For a CFA model with q factors, t typically exceeds $pq + p - q(q-1)/2$ because CFA imposes additional constraints (e.g., fixed loadings for identification).

2. **Applicability:** Bartlett's correction was derived specifically for EFA, where the factor loading matrix is rotationally indeterminate. Swain's correction is more general and applies to any covariance structure model, including CFA and SEM.

3. **Bias magnitude:** For models with the same p and q, Bartlett's correction typically produces a smaller adjustment than Swain's correction, because $2p + 4q + 1 < 2t + 2p + 3$ when $t > 2q$ (which is usually the case for CFA models with cross-loadings or correlated errors).

6.4 Yuan Empirical Correction (2015)

Yuan's Empirical Correction

Yuan (2015) proposed an empirical correction that adjusts the degrees of freedom rather than the test statistic:

$$T_{\text{Yuan}} = T_{\text{ML}} \times \frac{\text{df}^*}{\text{df}}, \tag{39}$$

where the adjusted degrees of freedom are:

$$\text{df}^* = \text{df} - \frac{2t + 2p + 3}{2}. \tag{40}$$

Yuan's (2015) approach is motivated by the observation that the expected value of T_{ML} under the null is:

$$\mathbb{E}[T_{\text{ML}}] = \text{df} + \frac{2t + 2p + 3}{2n} + O(n^{-2}), \tag{41}$$

If we define the adjusted degrees of freedom as $\text{df}^* = \text{df} - (2t + 2p + 3)/2$, then:

$$\mathbb{E}\left[\frac{T_{\text{ML}} \times \text{df}^*/\text{df}}{\text{df}^*/\text{df}}\right] \approx \text{df}^* \cdot \left(1 + \frac{2t + 2p + 3}{2n \cdot \text{df}}\right) + \frac{2t + 2p + 3}{2n} \cdot \frac{\text{df}^*}{\text{df}}. \tag{42}$$

A simpler interpretation is that Yuan's correction effectively "removes" the bias from the expected value by scaling the statistic down so that it is compared against a smaller effective degrees of freedom. The practical effect is that the corrected statistic is:

$$T_{\text{Yuan}} = T_{\text{ML}} \cdot \frac{\text{df} - (2t + 2p + 3)/2}{\text{df}}. \tag{43}$$

Since $(2t + 2p + 3)/2 > 0$, the scaling factor is less than 1, so the corrected statistic is smaller than the uncorrected statistic. This is consistent with the direction of the bias correction (the uncorrected statistic is too large).

Why Yuan's correction works under non-normality. A significant advantage of Yuan's empirical correction over the classical Bartlett and Swain corrections is its robustness to violations of the normality assumption. The classical corrections are derived under the assumption that the data are multivariate normal, and their theoretical guarantees (in particular, the cumulant matching property of Lawley's theorem) rely on this assumption. When the data are non-normal, the fourth-order cumulants of the sample covariance matrix deviate from their Wishart values, and the bias term ε may be either larger or smaller than the normal-theory value.

Yuan's correction addresses this issue by using the same bias formula $(2t+2p+3)/2$ regardless of the distribution of the data. While this might seem like a disadvantage (since the formula is derived under normality), Yuan (2015) showed that the correction actually provides *conservative* Type I error rates under non-normality. The reason is that non-normality tends to inflate the test statistic (Mardia's measure of multivariate kurtosis contributes positively to the expected value of T_{ML}), and Yuan's correction, by reducing the effective degrees of freedom, partially counteracts this inflation. The result is a test that is slightly conservative under non-normality but substantially more accurate than the uncorrected T_{ML}.

Relationship to Swain's correction. Yuan's correction and Swain's correction are closely related. Both use the bias term $(2t + 2p + 3)/2$ but apply it differently:

- Swain divides by $1 + (2t + 2p + 3)/(2n \cdot \mathrm{df})$, which is a first-order multiplicative correction to the statistic.
- Yuan multiplies by $(\mathrm{df} - (2t + 2p + 3)/2)/\mathrm{df}$, which is a first-order additive correction to the degrees of freedom.

To see the relationship, note that for large n:

$$\frac{1}{1 + (2t + 2p + 3)/(2n \cdot \mathrm{df})} \approx 1 - \frac{2t + 2p + 3}{2n \cdot \mathrm{df}}, \tag{44}$$

while:

$$\frac{\mathrm{df} - (2t + 2p + 3)/2}{\mathrm{df}} = 1 - \frac{2t + 2p + 3}{2\,\mathrm{df}}. \tag{45}$$

The key difference is that Swain's correction involves the sample size n in the denominator, while Yuan's correction does not. This means that Yuan's correction provides a fixed adjustment that does not diminish as n increases, which is advantageous for non-normal data where the bias may not decrease as rapidly with n as under normality. However, it also means that Yuan's correction may over-correct for large samples under normality, leading to conservative Type I error rates.

7 Bradley's Robustness Criterion

7.1 Definition and Motivation

In evaluating the performance of statistical tests, it is essential to assess not only whether the test achieves the nominal Type I error rate on average, but also whether the empirical rejection rate remains within acceptable bounds across different data conditions. Bradley (1978) proposed a widely used criterion for assessing the robustness of a statistical test with respect to its Type I error rate.

Bradley's Liberal Robustness Criterion

A statistical test operating at nominal significance level α is considered *robust* in the liberal sense if the empirical Type I error rate $\hat{\alpha}$ falls within the interval:

$$\left[\frac{\alpha}{2}, \frac{3\alpha}{2}\right]. \tag{46}$$

For $\alpha = 0.05$, this interval is $[0.025,\ 0.075]$.

The rationale for this criterion is that a test should not reject too rarely (which would indicate excessive conservatism and reduced power) or too frequently (which would indicate a failure to control the Type I error rate). The lower bound $\alpha/2$ ensures that the test maintains at least half of its nominal power, while the upper bound $3\alpha/2$ allows for a modest degree of inflation due to finite-sample effects and minor violations of the model assumptions.

7.2 Classification of Test Performance

Based on Bradley's criterion, we classify the performance of each correction method into three categories:

Table 2: Classification of test performance based on Bradley's criterion ($\alpha = 0.05$).

Category	**Empirical $\hat{\alpha}$**	**Interpretation**
Robust	$0.025 \leq \hat{\alpha} \leq 0.075$	Test maintains acceptable Type I error control
Liberal	$\hat{\alpha} > 0.075$	Test rejects too often; inflated Type I error
Conservative	$\hat{\alpha} < 0.025$	Test rejects too rarely; loss of power

A *robust* test satisfies Bradley's liberal criterion and is considered suitable for practical use. A *liberal* test has an inflated Type I error rate, meaning that the null hypothesis is rejected more often than the nominal level would suggest. This is particularly problematic in confirmatory research, where a significant result is taken as evidence against the null hypothesis. A *conservative* test has a deflated Type I error rate, meaning that it fails to detect true departures from the null hypothesis (reduced power). While conservatism is generally considered less serious than liberality, excessively conservative tests can lead to a failure to detect meaningful model misspecifications.

7.3 Practical Application

In our simulation study, we apply Bradley's criterion to each combination of correction method, model complexity, sample size, and data distribution. The empirical Type I error rate is estimated as the proportion of replications (out of $R = 5000$) in which the null hypothesis is rejected at the $\alpha = 0.05$ level. Under the normal approximation to the binomial distribution, the standard error of the estimated rejection rate is:

$$\mathrm{SE}(\hat{\alpha}) = \sqrt{\frac{\alpha(1-\alpha)}{R}} \approx \sqrt{\frac{0.05 \times 0.95}{5000}} \approx 0.0031. \tag{47}$$

This means that the 95% confidence interval for the empirical rejection rate under the null is approximately $\hat{\alpha} \pm 0.006$, providing sufficient precision to reliably classify each test as robust, liberal, or conservative.

8 Non-Normality: Impact on Bartlett Corrections

8.1 Why Bartlett Corrections Assume Normality

All classical Bartlett-type corrections, including those of Bartlett (1950) and Swain (1975), are derived under the assumption that the observed variables follow a multivariate normal distribution. This assumption enters the derivation at several critical points:

1. **Distribution of the sample covariance matrix.** The Wishart distribution, which underlies the derivation of the bias term ε, holds only under multivariate normality. Under non-normality, the sampling distribution of $\mathbf{S}$ is more complex, and the fourth-order cumulants Λ_{rstu} do not reduce to the simple form given in Equation (20).

2. **Form of the likelihood function.** The ML fitting function F_{ML} is derived from the multivariate normal log-likelihood. Under non-normality, the correct likelihood function involves higher-order moments, and F_{ML} is no longer the appropriate discrepancy function.

3. **Edgeworth expansion.** Lawley's theorem on cumulant matching relies on the Edgeworth expansion of the LRT statistic under normality. Under non-normality, the expansion involves additional terms that depend on the skewness and kurtosis of the data, and the simple multiplicative Bartlett correction does not fully capture these terms.

As a consequence of these dependencies, the classical Bartlett corrections may not provide accurate Type I error control when the data are non-normal. In particular, the bias term ε may be underestimated (leading to an insufficient correction) or overestimated (leading to an excessive correction), depending on the direction and magnitude of the non-normality.

8.2 Measuring Non-Normality: Skewness and Kurtosis

Multivariate non-normality is typically characterized by Mardia's (1970) measures of multivariate skewness and kurtosis. Mardia's skewness is defined as:

$$\beta_{1,p} = \mathbb{E}\Big[(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\Big]^3, \tag{48}$$

which reduces to 0 under multivariate normality and is positive when the distribution is skewed. Mardia's kurtosis is:

$$\beta_{2,p} = \mathbb{E}\Big[(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\Big]^2, \tag{49}$$

which equals $p(p+2)$ under multivariate normality. The excess kurtosis is $\beta_{2,p} - p(p+2)$, which is positive for leptokurtic (heavy-tailed) distributions and negative for platykurtic (light-tailed) distributions.

8.3 Choice of Non-Normality: Skewness = 2, Kurtosis = 3

In our simulation study, we generate non-normal data with marginal skewness of 2 and marginal kurtosis (excess kurtosis) of 3. This choice is motivated by several considerations:

1. **Moderate-to-severe non-normality.** A marginal skewness of 2 represents a substantial departure from normality (the normal distribution has skewness 0), and a marginal excess kurtosis of 3 represents a severe departure (the normal distribution has excess kurtosis 0). These values are commonly encountered in real-world psychological and social science data, where Likert-scale items and other bounded response formats often exhibit considerable non-normality.

2. **Consistency with prior simulation studies.** Several influential simulation studies in the SEM literature, including those by Curran, West, and Finch (1996), Hu, Bentler, and Kano (1992), and Fouladi (2000), have used similar levels of non-normality to evaluate the robustness of fit indices and test statistics. Using comparable values allows our results to be directly compared with prior findings.

3. **Impact on the LRT statistic.** Under non-normality with positive excess kurtosis, the expected value of T_{ML} is inflated beyond its normal-theory value. The magnitude of the inflation is approximately proportional to the excess kurtosis, and for excess kurtosis of 3, the inflation can be substantial for small to moderate sample sizes. This provides a challenging test case for the Bartlett corrections.

To illustrate the severity of this non-normality, consider a univariate distribution with skewness 2 and excess kurtosis 3. Such a distribution has a pronounced right tail and heavier tails than the normal distribution. When extended to the multivariate case, the joint distribution exhibits complex dependencies between the marginal shapes and the correlation structure, making the finite-sample behavior of the LRT statistic difficult to predict without simulation.

8.4 Vale and Maurelli's Method (1983)

The standard method for generating non-normal data with specified marginal skewness and kurtosis in SEM simulation studies is the method of Vale and Maurelli (1983). This method uses the Fleishman (1978) power transformation to convert normally distributed variables into variables with the desired marginal skewness and kurtosis, while approximately preserving the target correlation structure.

The Fleishman power transformation is defined as:

$$Y = a + bZ + cZ^2 + dZ^3, \tag{50}$$

where $Z \sim N(0, 1)$ and a, b, c, d are constants chosen so that Y has mean 0, variance 1, and the desired skewness γ_1 and excess kurtosis γ_2. For target skewness $\gamma_1 = 2$ and excess kurtosis $\gamma_2 = 3$, the Fleishman coefficients are approximately:

$$a = 0, \quad b = 0.494, \quad c = 0.189, \quad d = 0.066. \tag{51}$$

Vale and Maurelli's method proceeds in three steps:

1. **Compute the intermediate correlation matrix.** Given the target correlation matrix $\mathbf{\Sigma}$ and the Fleishman coefficients, compute the intermediate ("underlying normal") correlation matrix $\mathbf{\Sigma}_Z$ using the formulae derived by Vale and Maurelli (1983), which account for the nonlinear relationship between Z and Y.

2. **Generate multivariate normal data.** Draw a random sample from $N_p(\mathbf{0}, \mathbf{\Sigma}_Z)$.

3. **Apply the Fleishman transformation.** Transform each variable independently using Equation (50) to obtain a sample with the desired marginal skewness and kurtosis and approximately the target correlation structure.

While Vale and Maurelli's method does not guarantee exact preservation of the target correlation structure (due to the nonlinear transformation), the approximation is sufficiently accurate for simulation purposes when the target correlations are moderate in magnitude. The method has become the de facto standard for generating non-normal data in SEM simulation studies and is implemented in popular simulation packages such as `simsem` in R.

9 Summary of Correction Formulas

For convenience, we collect all correction formulas in a single summary table. This table provides a quick reference for the four correction methods, their formulas, and the conditions under which each is applicable.

Table 3: Summary of correction methods for the ML test statistic in SEM.

Method	**Corrected Statistic**	**Key Parameters**
Uncorrected ML	$T_{\text{ML}} = (n-1) \cdot F_{\text{ML}}(\hat{\boldsymbol{\theta}})$	n, p, t
Swain (1975)	$T_{\text{Swain}} = T_{\text{ML}} \cdot \dfrac{2t+2p+3}{2n-2t-2p-1}$	n, p, t
Bartlett EFA (1950)	$T_{\text{Bart}} = T_{\text{ML}} \cdot \dfrac{2n-2}{2n-2p-4q-3}$	n, p, q
Yuan (2015)	$T_{\text{Yuan}} = T_{\text{ML}} \cdot \dfrac{\text{df} - (2t+2p+3)/2}{\text{df}}$	n, p, t, df

Several observations emerge from this summary. First, all three correction methods multiply the uncorrected statistic T_{ML} by a factor less than 1, consistent with the positive bias of the uncorrected statistic. Second, Swain's correction and Yuan's correction share the same bias term $(2t + 2p + 3)/2$ but apply it differently (through the statistic vs. through the degrees of freedom). Third, Bartlett's EFA correction is the only method that depends on the number of factors q rather than the total parameter count t, reflecting its origin in exploratory factor analysis. Fourth, Yuan's correction is the only method that does not involve n in the correction factor, which contributes to its robustness under non-normality.

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